

DriverPass Business Requirements Document

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System Components and Design

Purpose

DriverPass has contracted our consulting company to develop a centralized system that supports both the educational and administrative sides of its business. The company wants a web-based application where customers can purchase training packages, access learning materials, schedule driving lessons, and monitor their progress. At the same time, the system must allow employees to efficiently manage users, appointments, and business operations through secure role-based access.

System Background

DriverPass identified a gap in the current driver education market because many new drivers are not adequately prepared to pass DMV examinations. To improve student success, the company plans to combine instructor-led driving sessions with digital learning resources in a single service.

The proposed system will support multiple business functions, including customer registration, appointment scheduling, online coursework, practice examinations, instructor management, reporting, and account administration. Authorized DriverPass employees will review changes to DMV rules and publish revised practice questions and instructional materials through the system. The company prefers a cloud-hosted solution that minimizes maintenance while remaining accessible through the internet.

Objectives and Goals

- Provide an online portal for customers to enroll in DriverPass services.
- Manage lesson packages and allow inactive packages to be removed from customer selection.
- Maintain customer records, payment information, and pickup locations.
- Schedule two-hour driving sessions while assigning available instructors and vehicles.
- Check the customer, instructor, and vehicle calendars before accepting a two-hour lesson reservation.
- Allow office personnel to create and update appointments for customers.
- Enable customers to independently manage their appointments through their accounts.
- Deliver online instructional content and practice examinations that reflect current DMV standards.
- Record examination results, lesson completion status, instructor feedback, and customer progress.
- Produce operational reports that management can download for business use.
- Maintain an audit log identifying users responsible for creating, modifying, or canceling records.
- Support administrative functions such as password resets, account management, and permission control.
- Operate as a secure cloud-based system that is accessible from internet-connected computers and mobile devices.

Requirements

Nonfunctional Requirements

Performance Requirements

- The system shall display standard pages and account information within four seconds during normal operating conditions.
- The system shall return available lesson times and confirm a reservation within eight seconds.
- The cloud-hosted application shall maintain at least 99.5% monthly availability, excluding announced maintenance periods.
- The system shall save appointment changes immediately and display a confirmation after the database accepts the transaction.
- Planned maintenance shall be performed during low-usage hours, with advance notice provided to employees when service interruptions are expected.

Platform Constraints

- DriverPass will deliver the application through a web browser, so users will not need to install a separate program on their devices.
- All live transactions will be processed against the hosted database; browser-cached information will not be treated as the official record.
- The back end will use relational storage to connect accounts, packages, appointments, vehicles, instructors, lessons, and test results.
- The interface will support the two most recent versions of Chrome, Edge, Firefox, and Safari on commonly used computers and mobile devices.
- Reports saved as spreadsheet files may be opened offline, but changes made to downloaded copies will not update the DriverPass database.
- The hosting company will maintain the computing environment, storage capacity, backup schedule, and recovery infrastructure.
- Users must reconnect to the internet before submitting coursework, changing account information, or managing a reservation.

Accuracy and Precision

- Each account, appointment, payment, and test attempt will receive a system-generated identifier that remains unchanged if descriptive information is edited.
- Registration forms will check email, telephone, date, time, and payment fields for valid formats before processing the submission.
- Password verification will preserve capitalization, while username and email lookup will ignore capitalization.
- A reservation will be accepted only if the instructor, vehicle, customer, and required package hours remain available when the transaction is committed.
- Remaining instructional time will be calculated from the purchased package and the duration of lessons marked complete.

- Test attempts will be attached to the correct customer and will retain the examination name, submission time, result, and completion status.
- If a transaction fails, the system will cancel the incomplete operation, preserve the previous record, and create a diagnostic event for IT review.

Adaptability

- User permissions will be stored as configurable role settings so access rules can be adjusted without rebuilding the application.
- The administrative console will allow authorized IT personnel to change account status, role assignment, and credential-recovery settings.
- Course revisions will move through draft, review, and publication stages so employees can update learning content independently of a software deployment.
- Existing packages will include effective-status controls that management can use to determine whether customers may select them.
- Application corrections will be deployed on the hosted server and will become available when the user reloads the browser.
- Compatibility testing will be performed after major browser or mobile operating-system releases.
- New package structures or changes to included modules will remain outside the configurable options in the first release.

Security

- Protected pages will require an authenticated DriverPass account.
- Permissions attached to the account will determine which menus, records, and actions become available after sign-in.
- Browser traffic will be protected through an encrypted HTTPS connection.
- Passwords will be converted to salted, nonreversible hashes before storage.
- After three unsuccessful sign-in attempts, the system will introduce progressively longer delays. Ten consecutive failures will suspend further attempts until the account is verified or reviewed by IT.
- The system will display an expiration warning before closing a session that has been inactive for 20 minutes.
- Password recovery will issue a single-use link to the account's verified email address, and the link will expire after 30 minutes.
- Customer, transaction, and backup data will be encrypted wherever sensitive information is retained.
- Security records will capture failed authentication, permission changes, account suspensions, and other administrative actions.

Functional Requirements

<i>Functional area</i>	<i>Requirement</i>
<i>Account creation</i>	The system shall allow a customer to establish an account with a unique username and password.
<i>Customer profile</i>	The system shall maintain the customer's name, address, telephone number, email address, and preferred pickup and drop-off locations.
<i>Package catalog</i>	The system shall present active training packages with their prices, services, instructional materials, and included driving hours.
<i>Package enrollment</i>	The system shall allow a customer to select one available training package for enrollment.
<i>Package purchase</i>	The system shall submit the information required to authorize payment and record the resulting transaction with the customer's enrollment.
<i>Learning access</i>	The system shall open only the lessons, practice examinations, and other resources included with the customer's package.
<i>Examination processing</i>	The system shall score a submitted practice examination and retain its name, submission time, score, and completion status.
<i>Training summary</i>	The system shall show the customer's completed lessons, examination results, instructor observations, and unused training hours.
<i>Availability search</i>	The system shall present open two-hour driving appointments based on recorded instructor and vehicle availability.

<i>Functional area</i>	<i>Requirement</i>
<i>Customer scheduling</i>	The system shall allow customers to reserve, move, or cancel their own driving appointments through the online calendar.
<i>Office scheduling</i>	The system shall let authorized office employees manage reservations for customers who contact DriverPass by telephone or in person.
<i>Reservation record</i>	The system shall connect every accepted appointment with one customer, one instructor, one vehicle, a two-hour time period, and pickup and drop-off information.
<i>Resource validation</i>	The system shall decline a reservation when any part of the requested time conflicts with another assignment for the instructor or vehicle.
<i>Package-hour validation</i>	The system shall decline a reservation when the customer lacks enough unused instructional time for the requested lesson.
<i>Instructor itinerary</i>	The system shall present each instructor with a daily list of assigned customers, appointment times, and pickup locations.
<i>Lesson completion</i>	The system shall provide an end-of-lesson form for recording completion, actual duration, and instructor observations.
<i>Hour balance</i>	The system shall reduce the customer's available package hours after a lesson is marked complete.
<i>Customer lookup</i>	The system shall let approved employees search for customers by account number, name, email address, or telephone number and save authorized profile changes.

Functional area	Requirement
Customer communication	The system shall provide authorized employees with the customer contact information needed for appointment and training communication.
Package control	The system shall let management determine whether an existing package is available for new enrollment.
Content workflow	The system shall maintain draft and published versions of instructional material while authorized employees review revisions.
Content release	The system shall release an approved course revision only to customers whose enrolled packages include the affected material.
Reporting	The system shall let the owner filter, view, print, and download reports concerning customers, reservations, instructional activity, and business operations.
Permission control	The system shall construct each user's navigation and record access from the permissions assigned to that account.
Account administration	The system shall provide IT personnel with account-lifecycle controls covering activation, access suspension, credential recovery, and role assignment.
Password assistance	The system shall allow a user to begin an automated password-recovery process through the verified email address attached to the account.
Audit history	The system shall preserve the responsible user, time, previous value, and resulting value when significant account or appointment information changes.

User Interface

- The sign-in screen will contain credential fields, password-recovery access, and messages explaining unsuccessful authentication.
- After login, the home screen will show only the navigation options permitted for that account.
- The enrollment screen will present active packages, included services, instructional hours, and purchase controls.
- The learning center will organize available lessons, practice examinations, prior attempts, and progress information.
- The appointment calendar will allow users to filter open two-hour reservations by date and availability.
- A reservation-review screen will display the customer, instructor, vehicle, time, and pickup information before the appointment is submitted.
- The instructor lesson screen will provide the scheduled customer information and fields for completion status, actual times, and observations.
- The employee customer-search screen will support registration, profile updates, package enrollment, scheduling, and customer contact.
- The content editor will separate drafts from published material and provide review and publication controls.
- The reporting screen will include date filters, report previews, printing, and spreadsheet export.
- The administration screen will provide account-status, permission, role, recovery, and activity-log controls.
- Invalid fields will be identified beside the affected entry, while correctly entered information will remain in the form.
- Completed transactions will return a confirmation containing the record number and updated status.

Assumptions

- The initial deployment will support the ten DriverPass training vehicles and their regular instructor assignments described during the interview.
- Driving appointments will continue to use two-hour scheduling blocks.
- DriverPass will approve the package definitions, prices, instructional hours, and content entitlements before configuration begins.

- DriverPass will retain responsibility for reviewing DMV changes and approving educational revisions.
- The selected payment and email providers will supply interfaces that the application can use for transaction authorization and account recovery.
- Employees will complete role-specific training before receiving production access.
- The owner or another authorized client representative will approve the requirements and final system before release.

Limitations

- The first release will provide a responsive website rather than separate native applications for iOS and Android.
- Appointments will use fixed two-hour blocks; support for variable lesson lengths is outside the initial design.
- The project does not include an automated DMV data feed or responsibility for interpreting regulatory changes.
- Package availability may be changed by management, but administrators cannot redesign package contents in the first release.
- No process has been defined for importing historical paper records or existing spreadsheets.
- Final capacity estimates cannot be established until DriverPass supplies expected user volume and simultaneous-connection information.
- Service operation will depend on the availability of the selected hosting, payment, email, and internet providers.
- Features requested after customer approval may require additional development time, funding, and schedule changes.

Gantt Chart

DriverPass Project Schedule

